

SUPSI

Challenges related to the optical recognition of the eggs of four *Aedine* mosquito species



Nikoleta Anicic, Diego Parrondo, Damiana Ravasi & Eleonora Flacio,
Vector Ecology Unit, Institute of Microbiology, SUPSI, 6850 Mendrisio, Switzerland

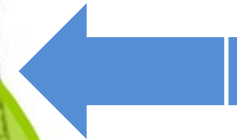
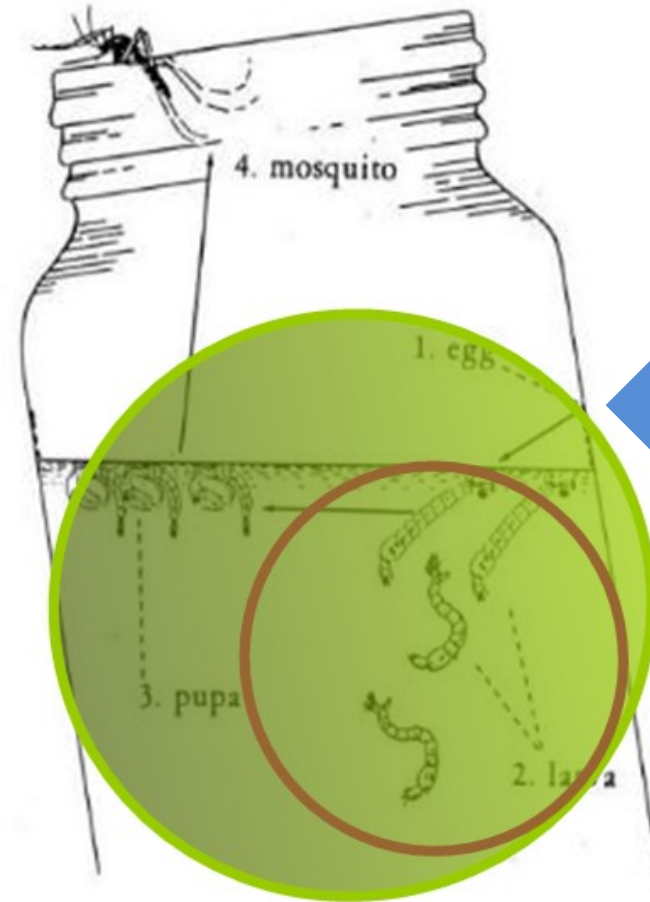
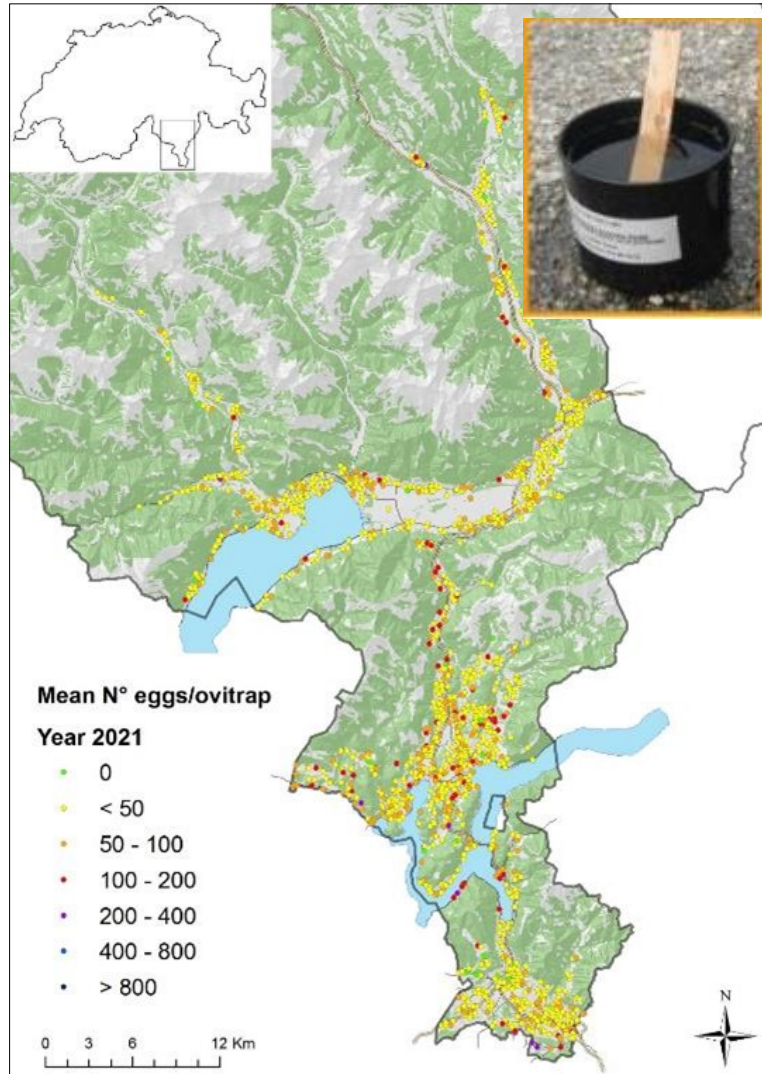
AE. ALBOPICTUS CANTONAL SURVEILLANCE AND CONTROL

SUPSI



reduce the risk of local transmission of diseases

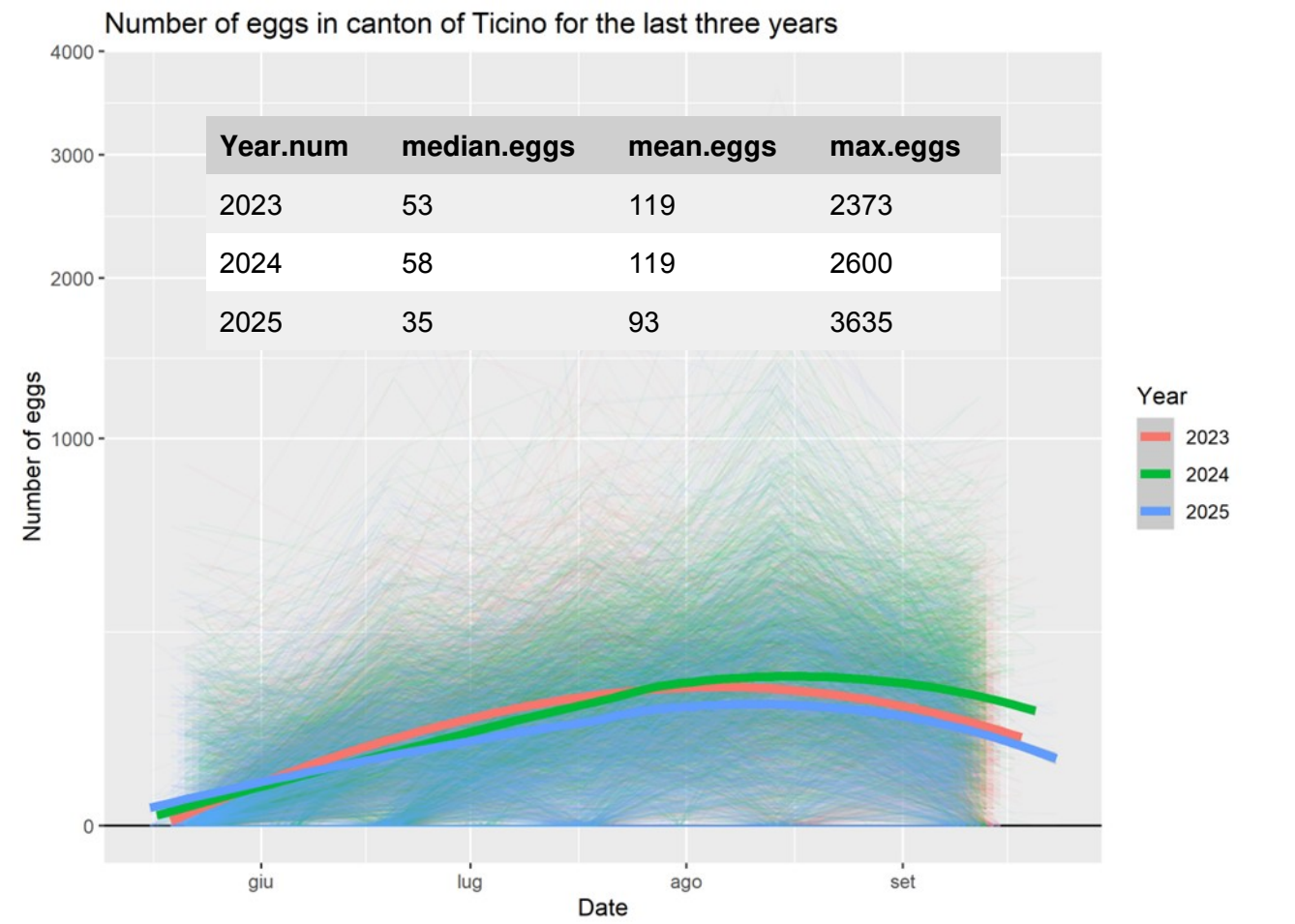
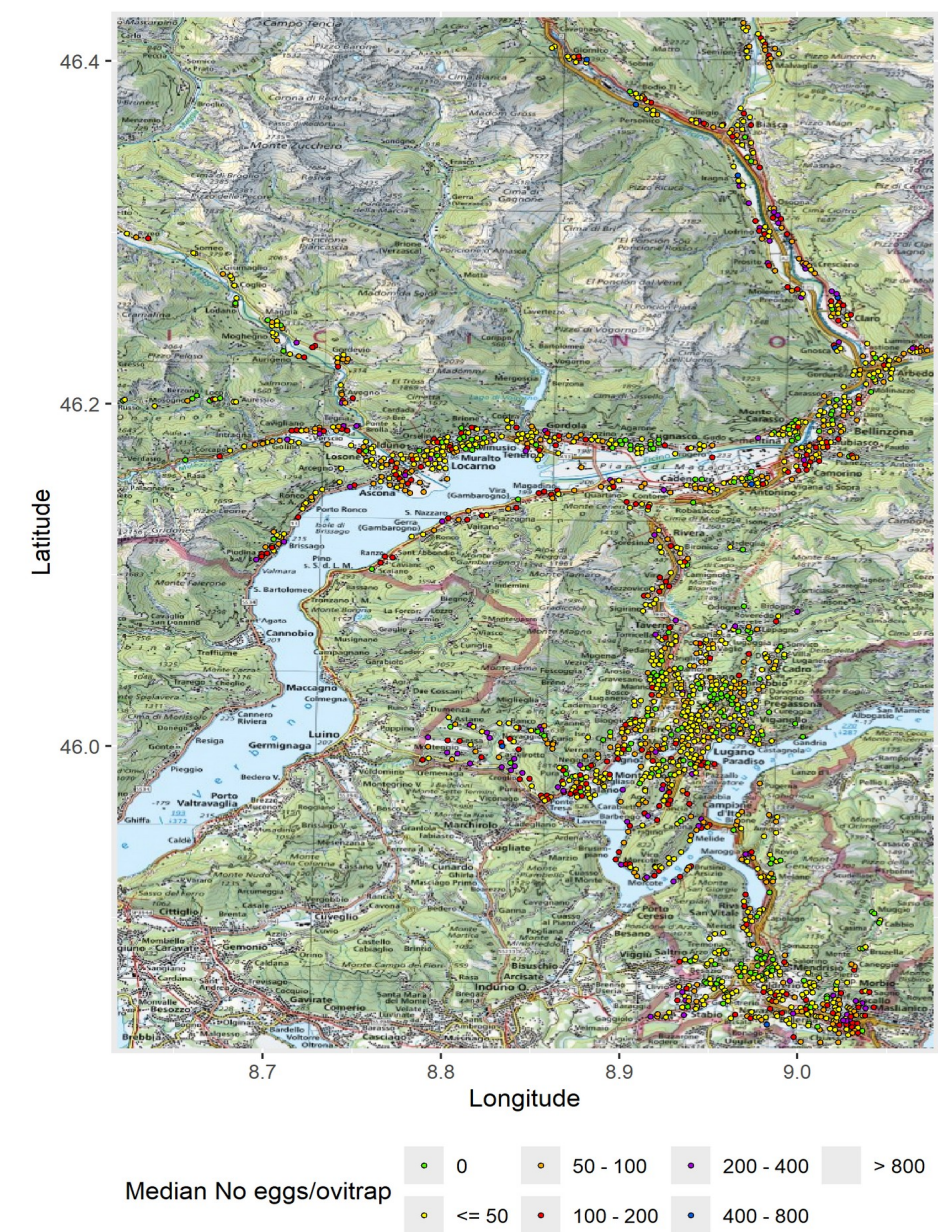
keep the nuisance at acceptable levels



ADDITIONAL
RESEARCH



RESULTS 2025 (Canton Ticino)



% difference in the number of eggs detected in 2025 vs 2022 and 2023

levels.year	median.eggs	mean.eggs	max.eggs
2023	-51	-13	35
2024	-66	-27	28

[Deutsch](#) [Français](#) [Italiano](#) [English](#)

 **Swiss Mosquito Network**

[About](#) [Invasive Mosquitoes](#) [Report an Invasive Mosquito](#) [News archive](#)

[Admin page](#)

Swiss Mosquito Network

Information on invasive mosquitoes in Switzerland and suggestions to prevent spread.

[Report an invasive mosquito](#)

www.muecken-schweiz.ch; www.moustiques-suisse.ch; www.zanzare-svizzera.ch; www.mosquitoes-switzerland.ch

News



SWISS MOSQUITO NETWORK

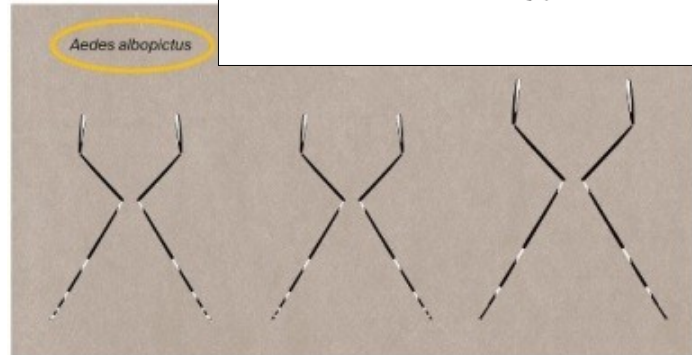
SUPSI

Swiss Mosquito Information on Switzerland and



How to Report a Tiger Mosquito

1. First observe well the insect:
 - It has two wings, relatively small (about 1.5 cm)
 - It has black and white striped legs
 - The thorax of *Aedes albopictus* is black brownish / yellowish
2. Take a full picture of the specimen corner



3. Continue your report by clicking the button below.

Report a Tiger Mosquito

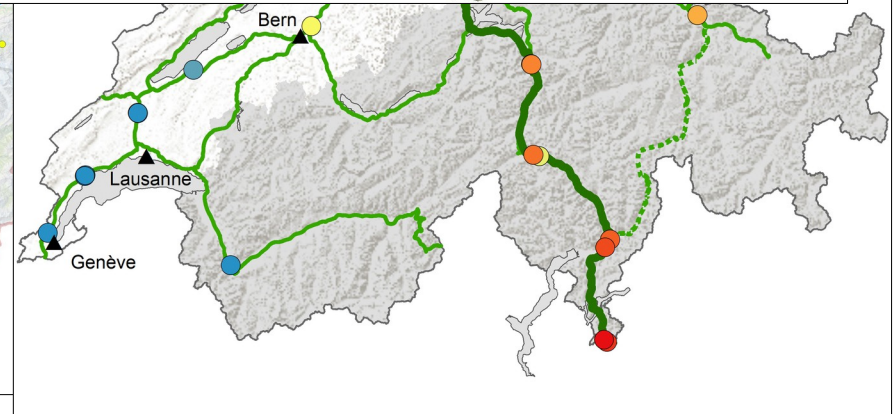
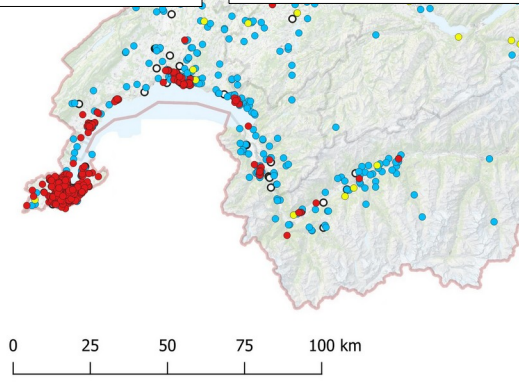
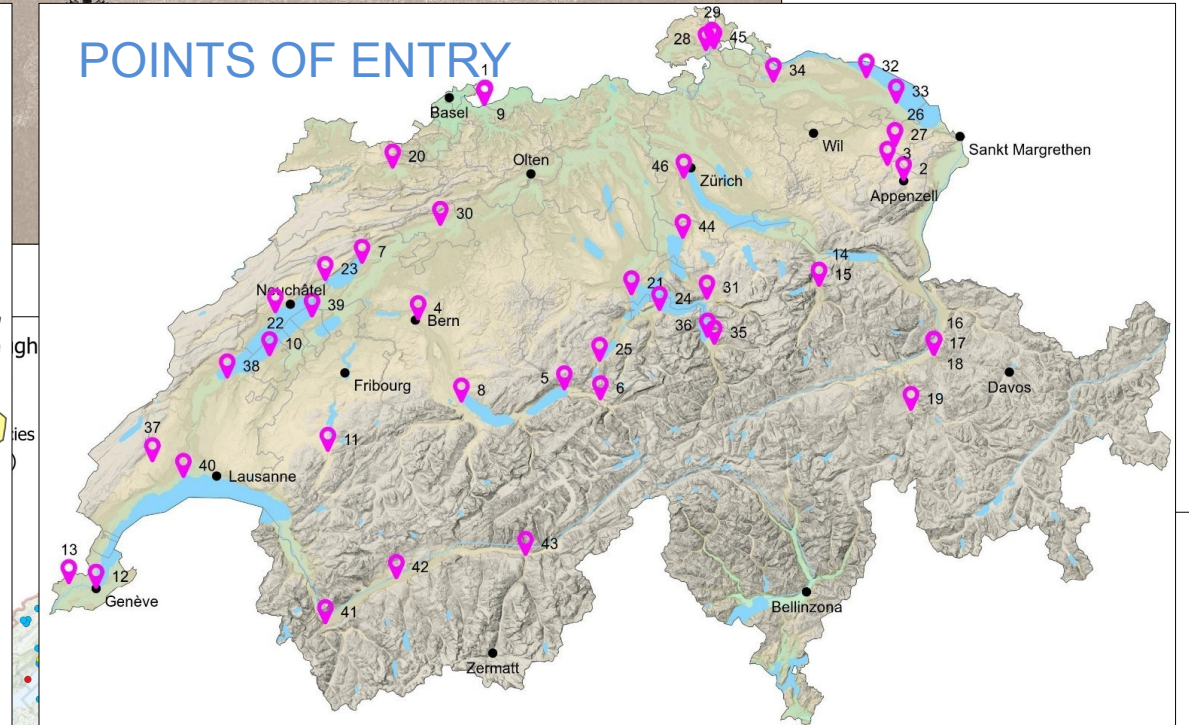
4. If possible, preserve the specimen (place it between two small pieces of paper towels and then store it in a small box) to prevent deterioration. If the photo is not sufficient for reliable identification, the reporting center will ask you to send the specimen.

CANTONS

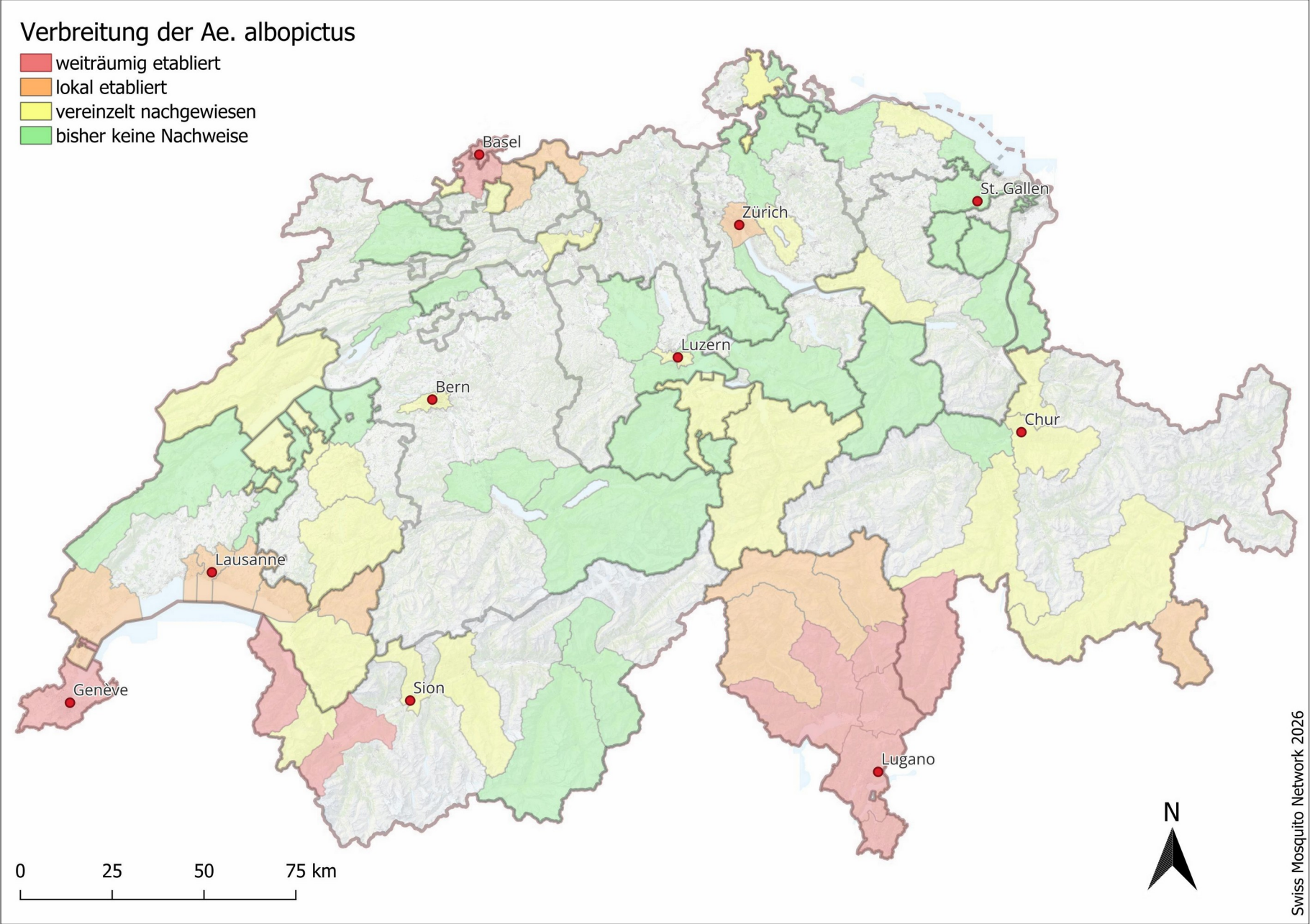
- 1 Zürich
- 2 Bern
- 3 Luzern
- 4 Uri
- 5 Schwyz
- 6 Obwalden
- 7 Nidwalden
- 8 Glarus
- 9 Zug
- 10 Fribourg / Freiburg
- 11 Solothurn
- 12 Basel-Stadt
- 13 Baselstid
- 14 Schaffhausen
- 15 Appenzell Ausserrhoden
- 16 Appenzell Innerrhoden
- 17 St. Gallen
- 18 Graubünden / Grischun / Grigioni
- 19 Aargau
- 20 Thurgau
- 21 Ticino / Tessin
- 22 Vaud / Waadt
- 23 Valais
- 24 Neuchâtel / Neuenburg
- 25 Genève / Genf
- 26 Jura



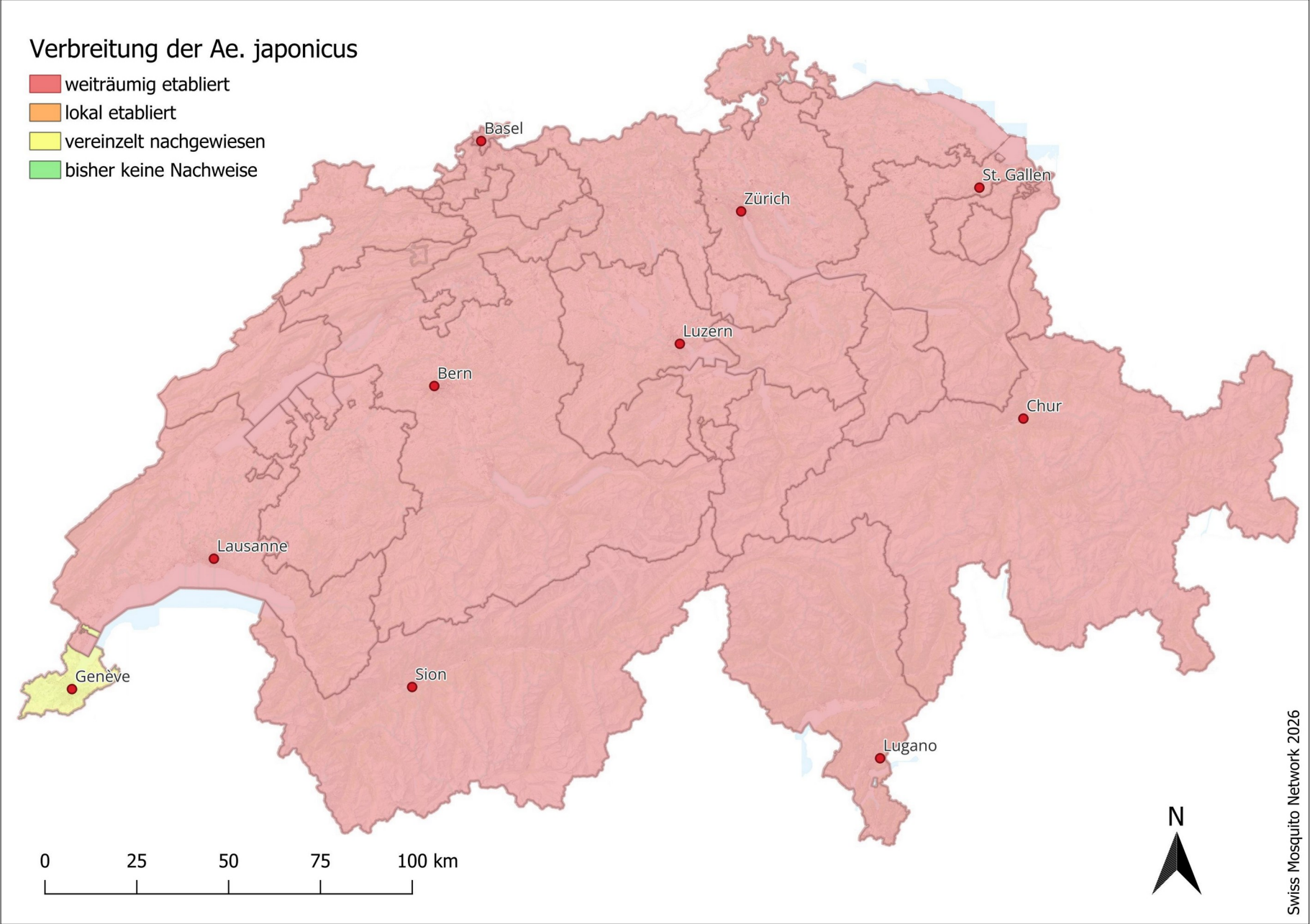
POINTS OF ENTRY



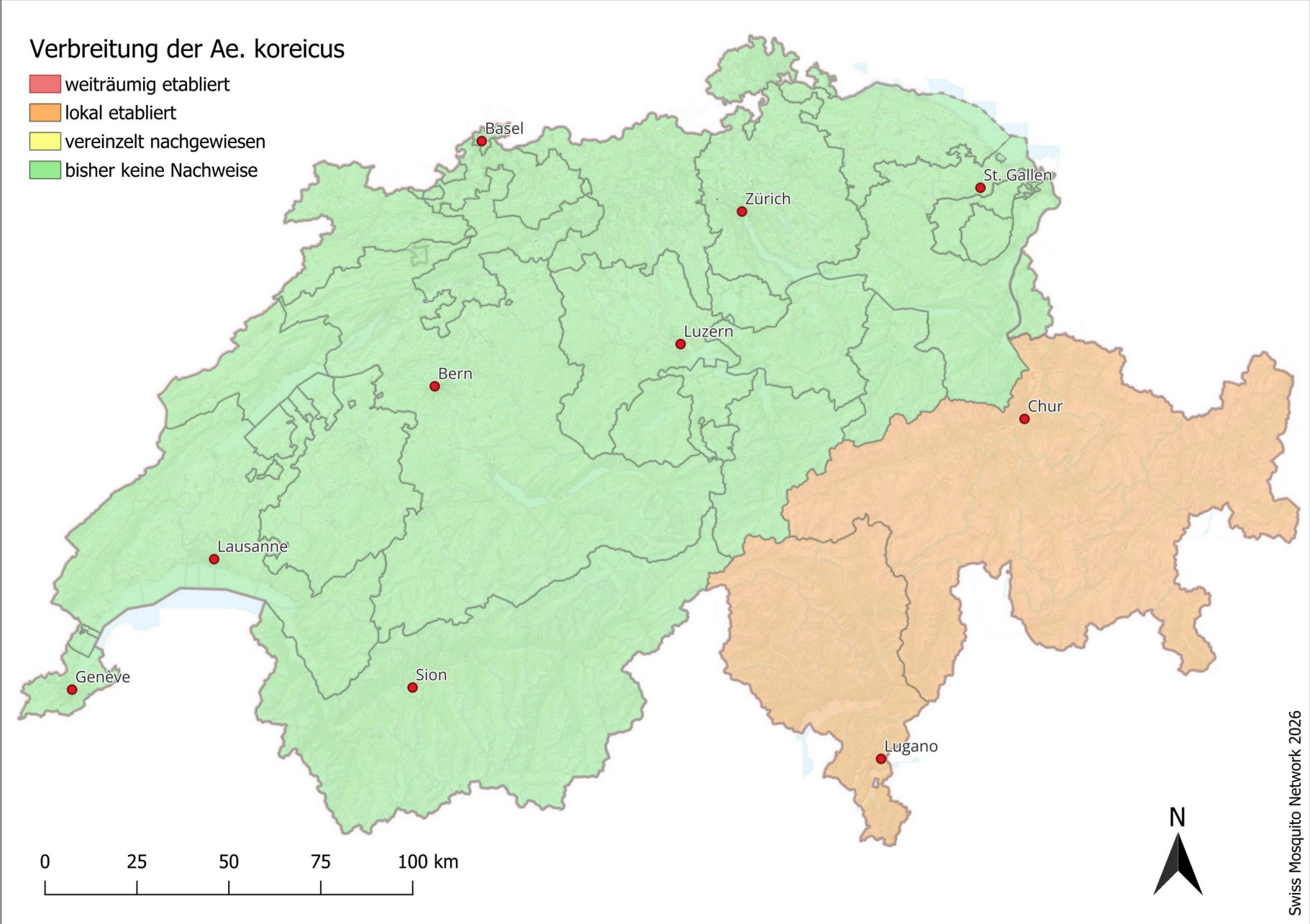
AEDES ALBOPICTUS IN SWITZERLAND: 2025



AEDES JAPONICUS IN SWITZERLAND: 2025

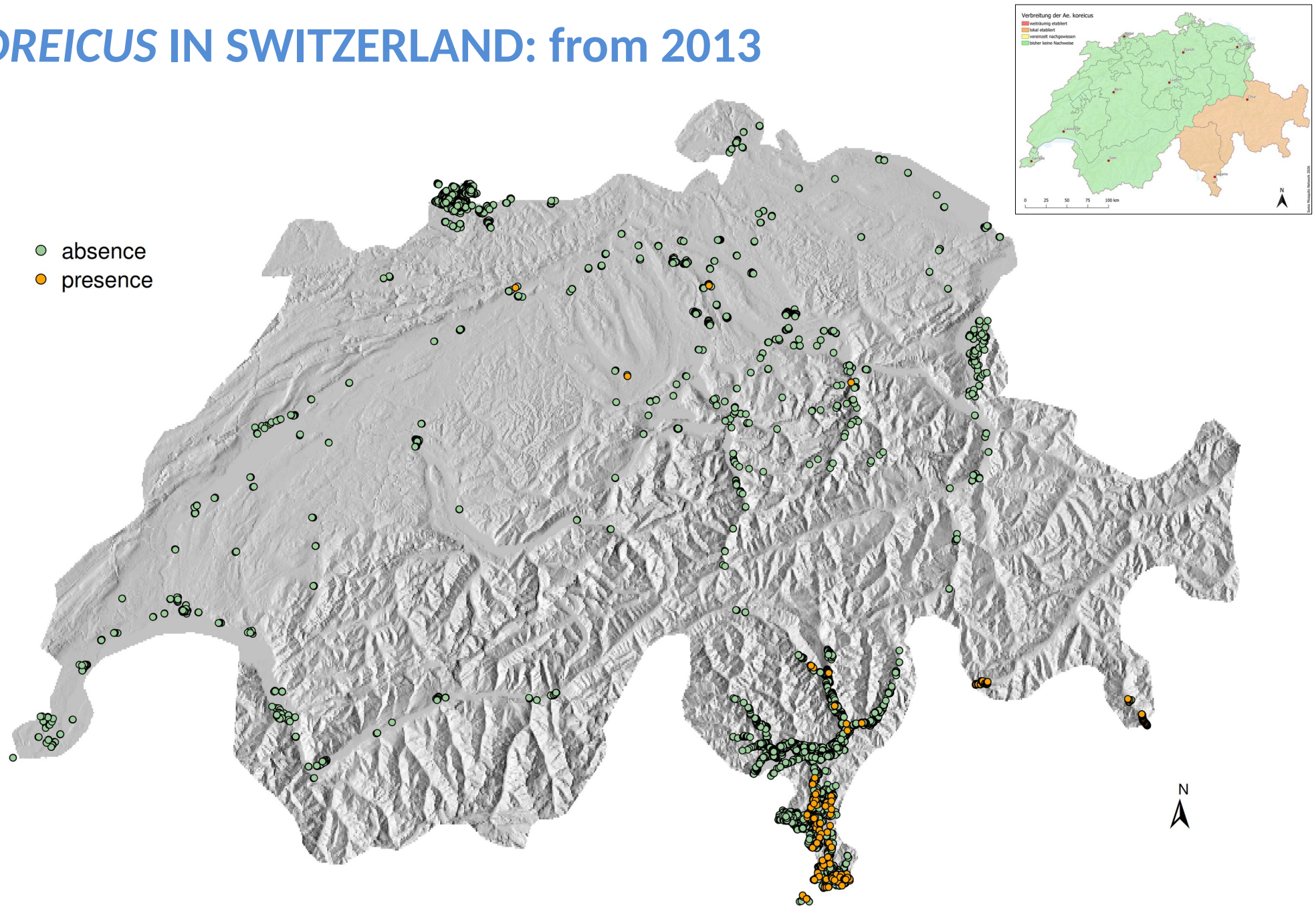


AEDES KOREICUS IN SWITZERLAND: 2025

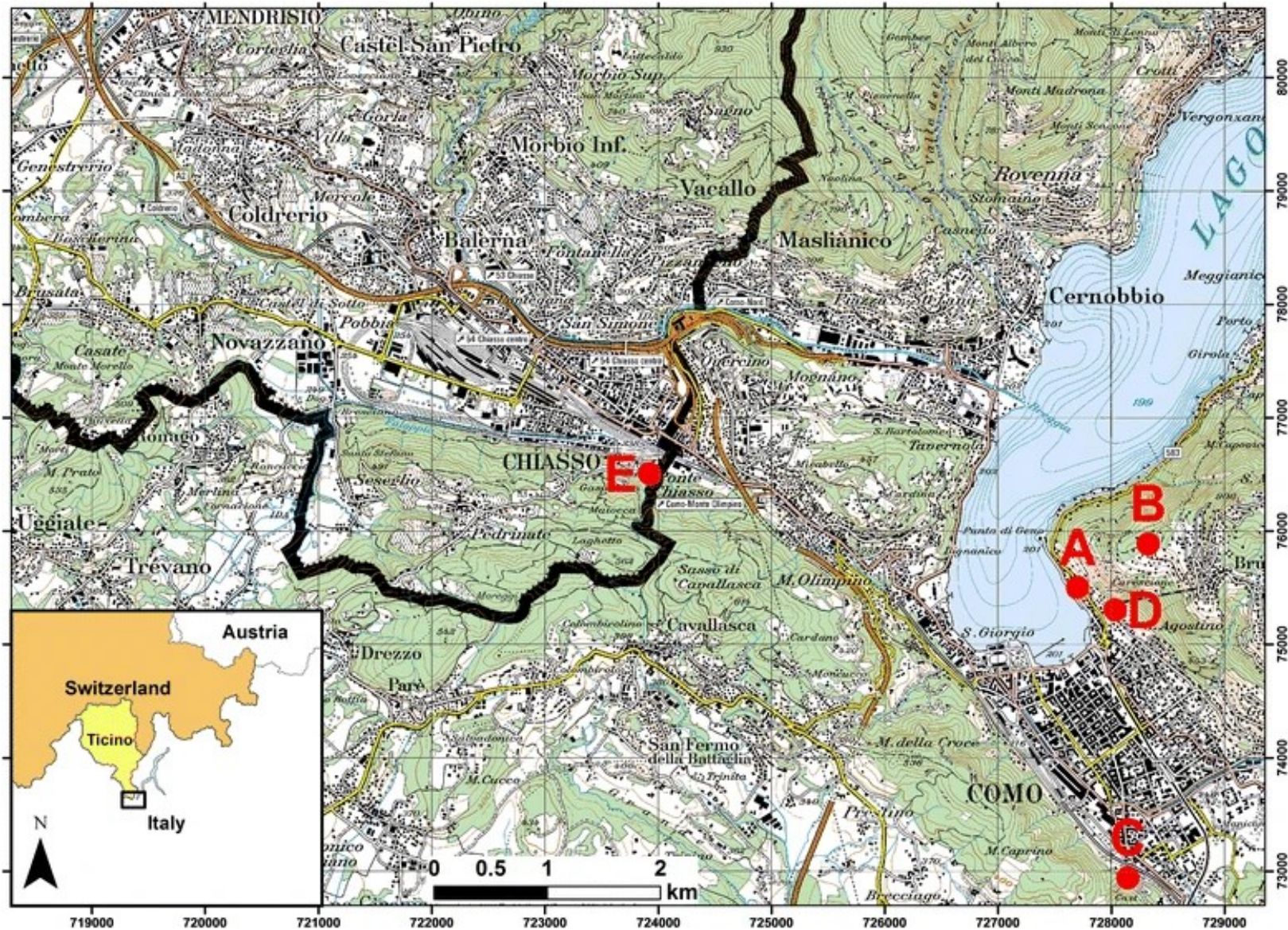


AEDES KOREICUS IN SWITZERLAND: from 2013

SUPSI



AEDES KOREICUS IN SWITZERLAND: 2013

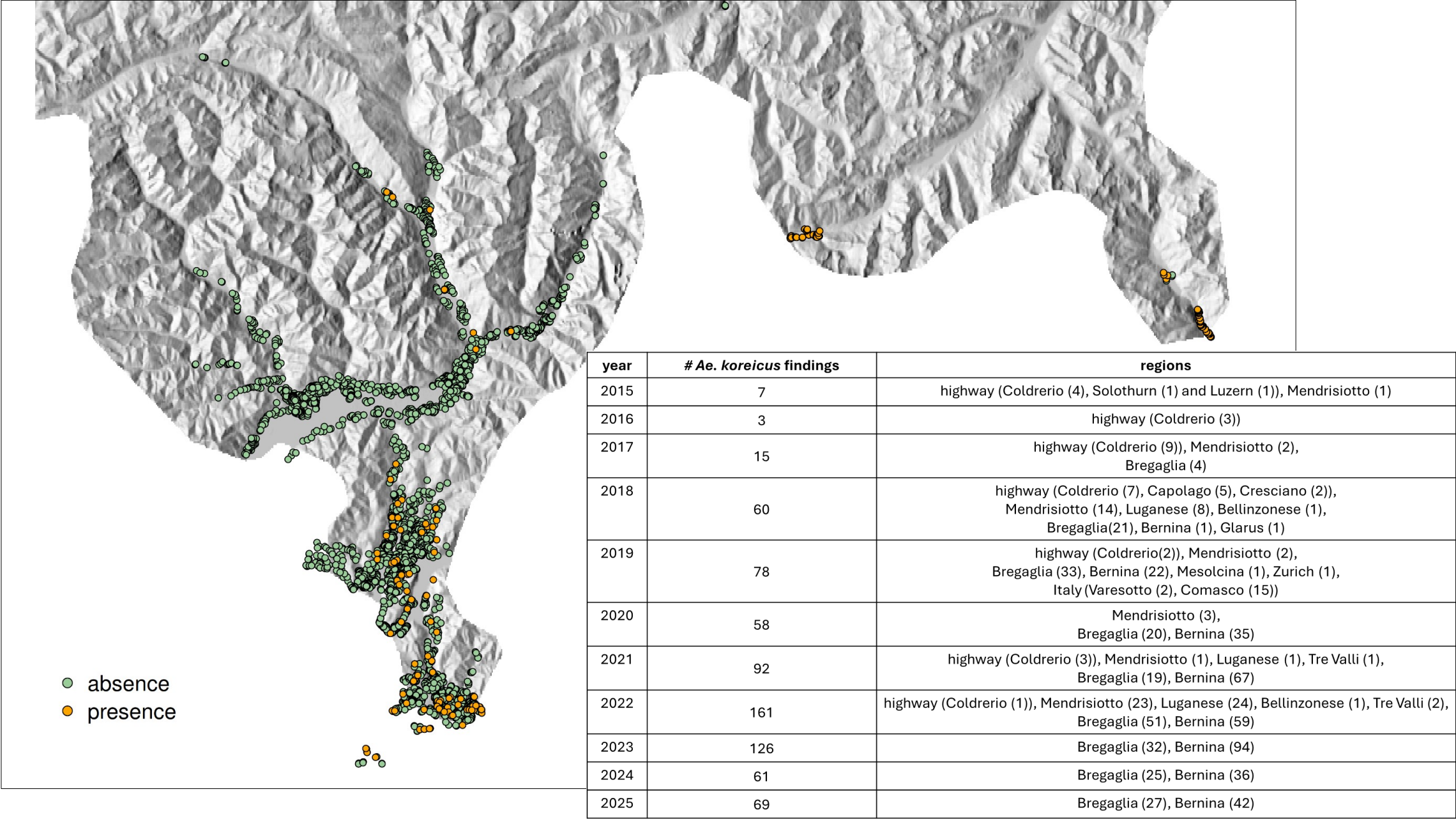


AEDES KOREICUS IN SOUTHERN SWITZERLAND

SUPSI

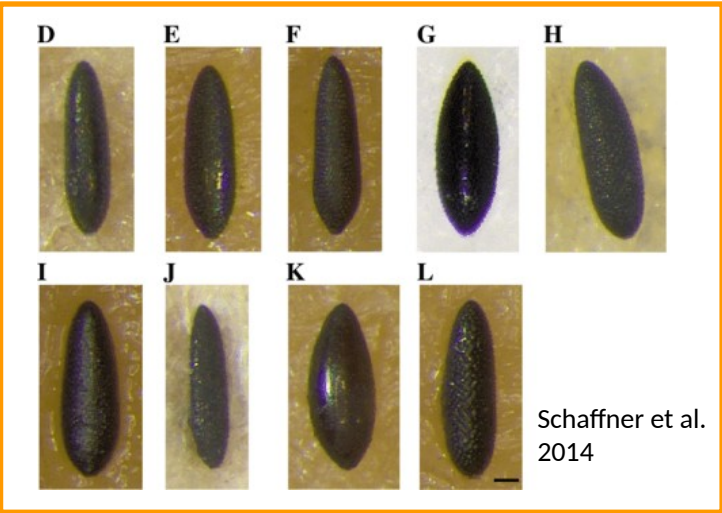


AEDES KOREICUS IN SOUTHERN SWITZERLAND: 214-1,120.9 m a.s.l.



ANALYSIS OF OVITRAPS

>1'500 ovitraps checked every 2 weeks (May-September)

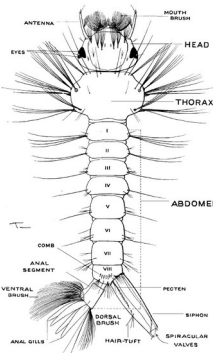


Schaffner et al. 2014

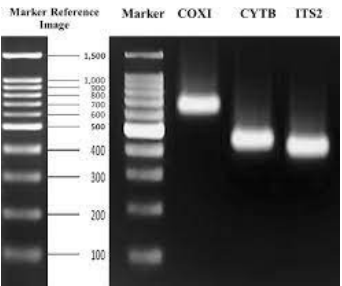
SPECIES DETERMINATION



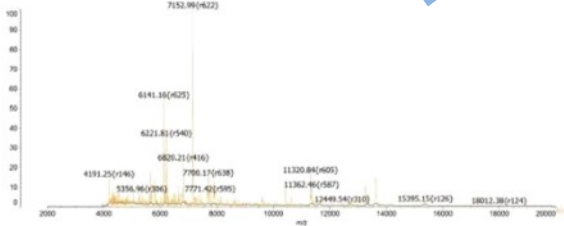
egg counting



hatching



PCR



MALDI-TOF MS



Database



EGGS IDENTIFICATION

- Presence: *Ae. albopictus*, *Ae. japonicus*, *Ae. koreicus* and *Ae. geniculatus*
- *Ae. aegypti*?
- Target species: *Ae. albopictus*.



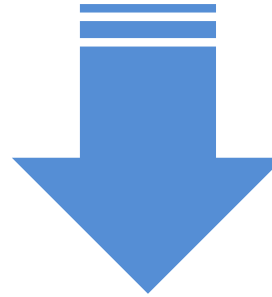
EGGS IDENTIFICATION: MALDI-TOF MS analyses

➤ Expensive

➤ Laborious

- they require compounds (formic acid and matrix)
- MS device
- time for the preparation: 1 hour for slide preparation (46 samples) + 30–60 min analysis

➤ The analyses that are done (budget-limited) are only a **small part of the total number of eggs** present on a wooden paddle



Find a method that is **faster**, **cheaper** and that gives us a **complete analysis** of the species of all eggs

EGGS IDENTIFICATION: chorion optical identification

- Preliminary tests (2018)
- MALDI-TOF MS to confirm the species



FIRST RESULTS	# of analysed eggs	# of correctly identified eggs	% of matches
<i>Aedes albopictus</i>	12	12	100
<i>Aedes japonicus</i>	18	17	94.44
<i>Aedes koreicus</i>	18	13	72.22
<i>Aedes geniculatus</i>	7	7	100
Total	55	49	89.09

EGGS IDENTIFICATION: chorion optical identification

- inter-rater reliability tests (questionnaires)
- *Ae. albopictus* and *Ae. geniculatus* > 86%
- *Ae. japonicus* and *Ae. koreicus* < 59%

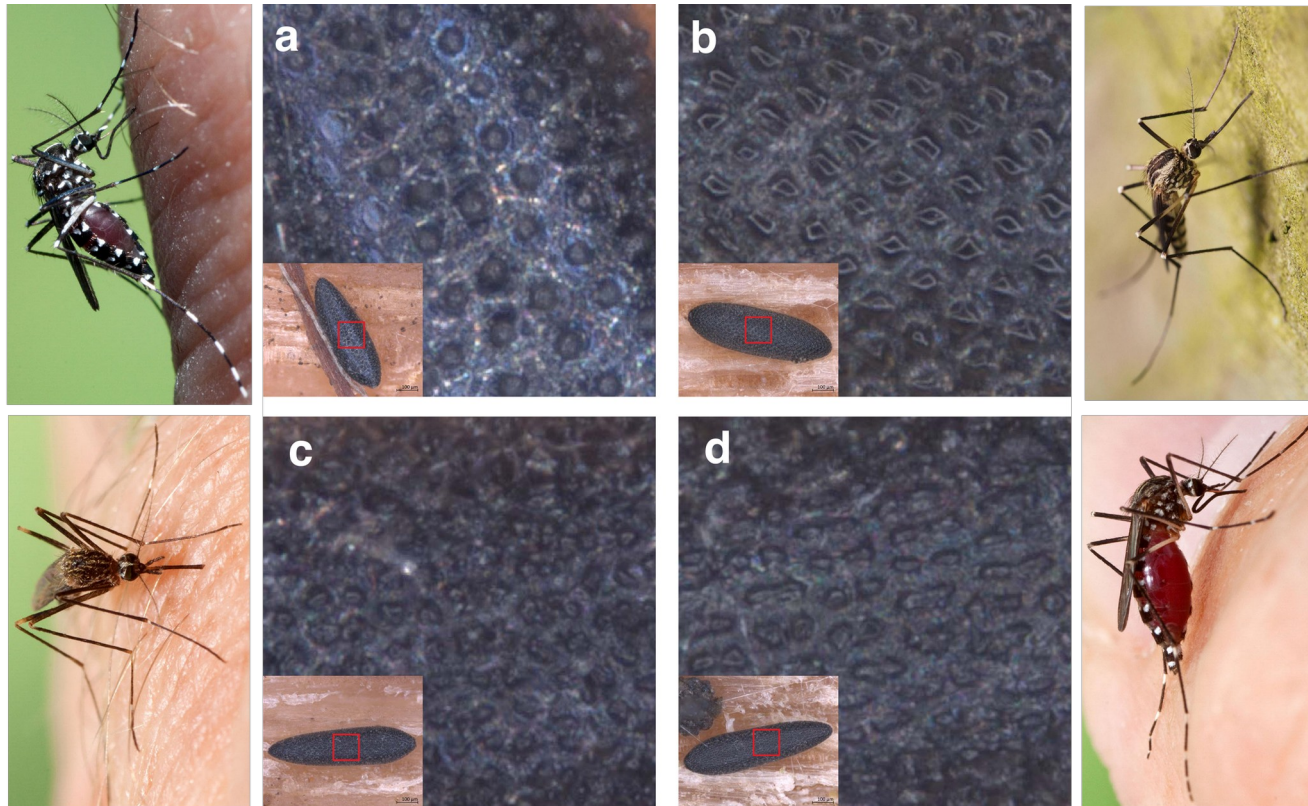
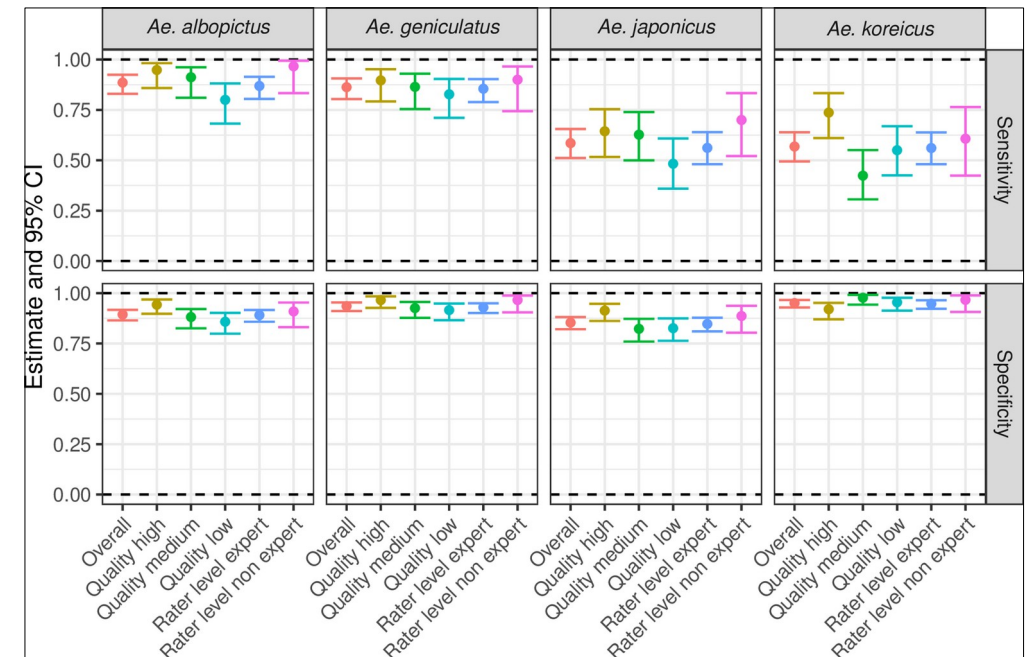


Fig 1. Patterns present on exochorion membrane. Egg of *Aedes albopictus* (a), *Aedes geniculatus* (b), *Aedes japonicus* (c), and *Aedes koreicus* (d).

Anicic Nikoleta et al. 2023. «Optical Recognition of the Eggs of Four Aedine Mosquito Species (*Aedes albopictus*, *Aedes geniculatus*, *Aedes japonicus*, and *Aedes koreicus*)». PLoS One 18 (11): e0293568. <https://doi.org/10.1371/journal.pone.0293568>

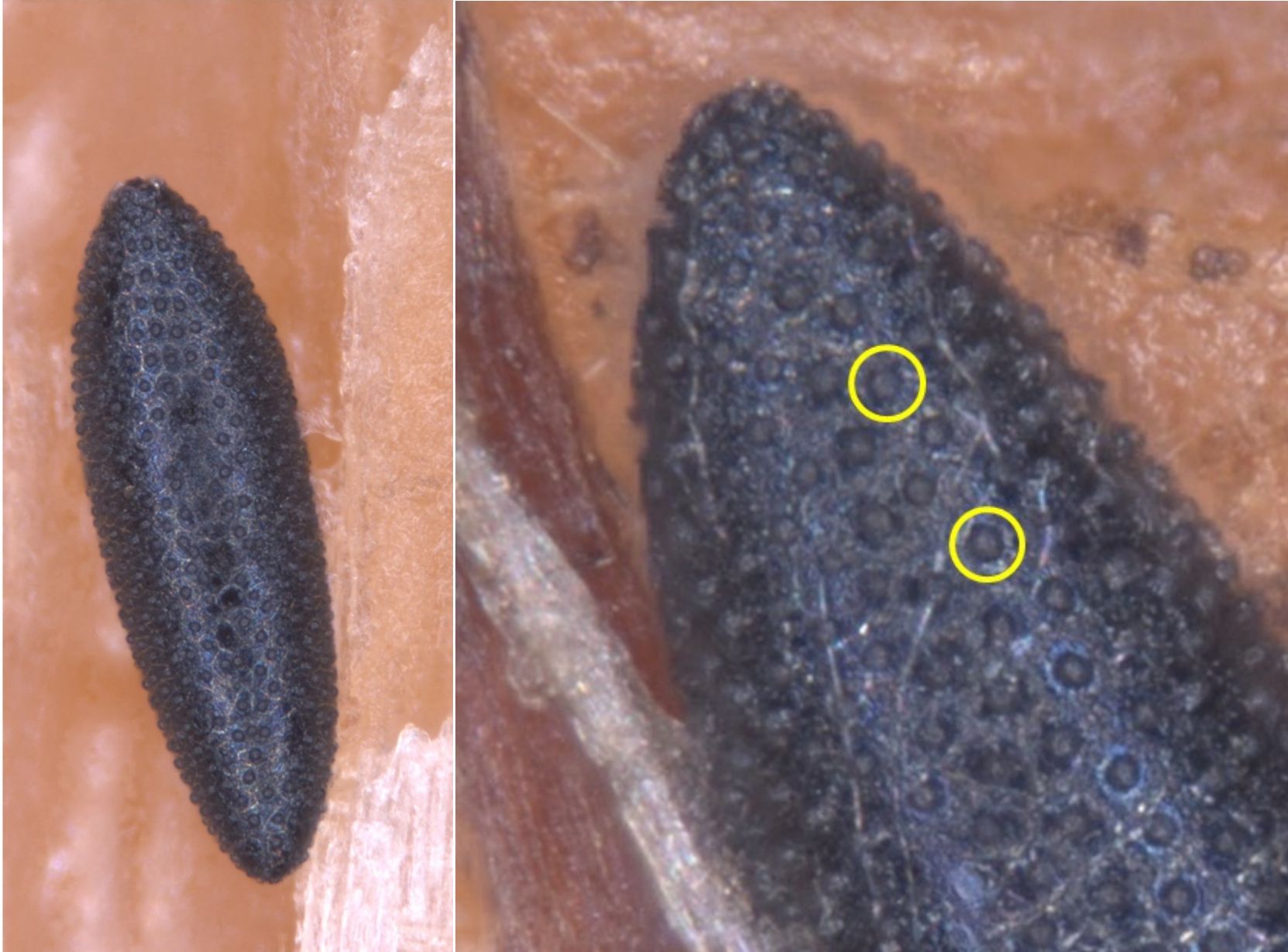
Table 1. Number of eggs optically analysed in relation-the species and the exochorion quality. The species was determined via MALDI-TOF.

Species	Exochorion quality			Total
	high quality	medium quality	low quality	
<i>Ae. albopictus</i>	23	29	10	62
<i>Ae. geniculatus</i>	16	8	8	32
<i>Ae. japonicus</i>	48	26	9	83
<i>Ae. koreicus</i>	18	19	23	60
Total	105	82	50	237



Aedes albopictus

SUPSI



The outer chorionic cell consist of **one large spherical central tubercle**.

It is also possible to recognize this structure with a large magnification binocular.

Aedes japonicus and *Aedes koreicus*



Ae. japonicus: the outer chorionic cell consist of a small group of 2-4 round tubercles

Ae. koreicus: the outer chorionic cell consist of a small group of 2-3 long tubercles.

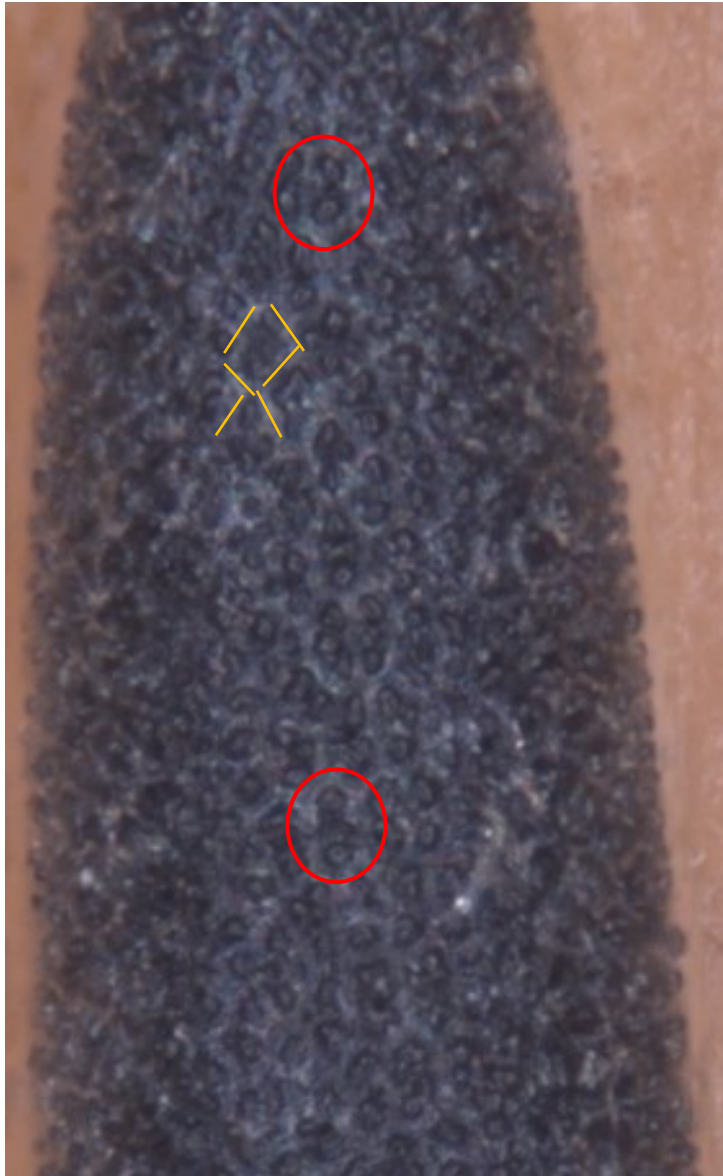
Given the close similarity between the two species, it is impossible to determine with certainty using a normal bionocular. Further analyses are required to reach a definitive determination.

The only visible difference lies in the length of the eggs.

Ae. koreicus appears to be longer, however, this characteristic is not a reliable method of determination.

Aedes japonicus

SUPSI

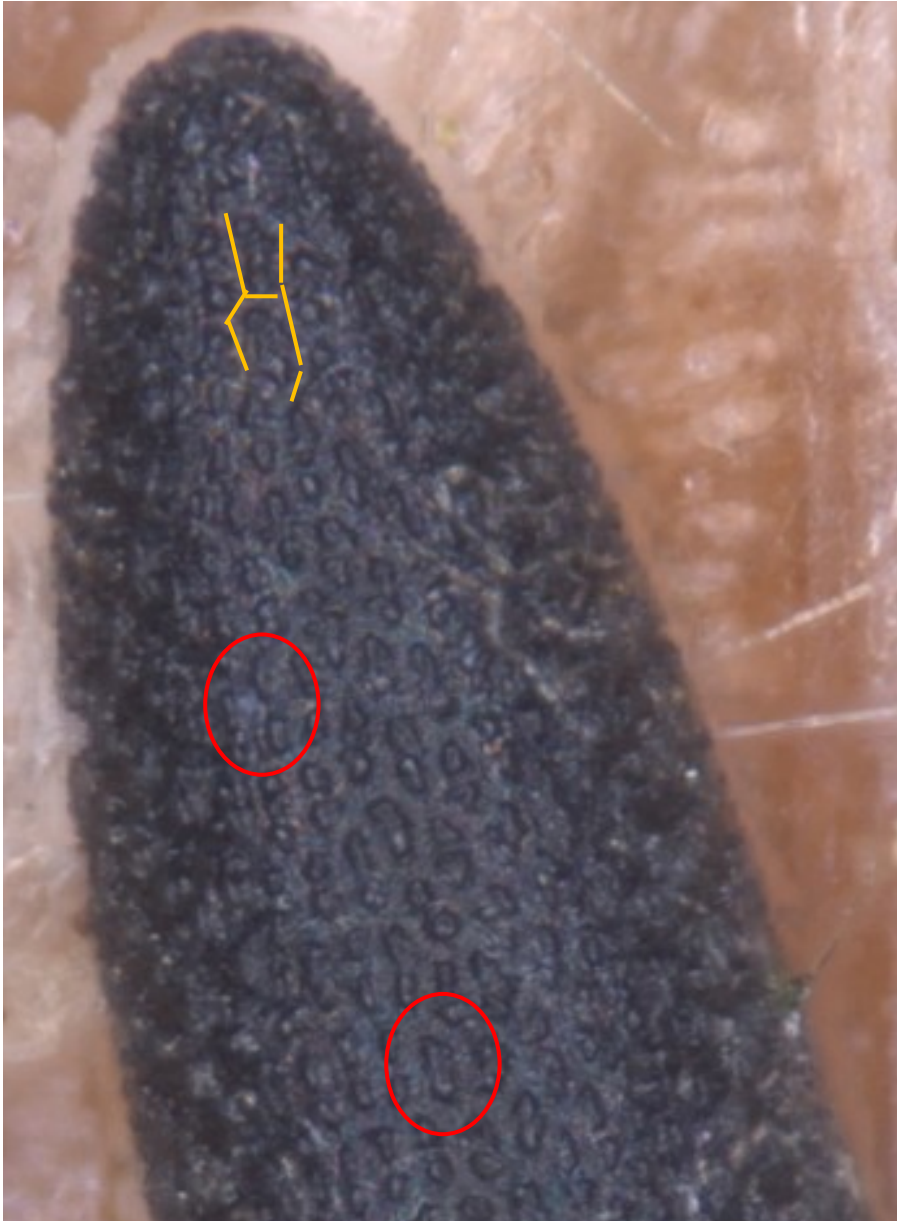


The **outer chorionic cell** consist of a small group of 2-4 round tubercles.

The **exochorionic network** that is found between outer chorionic cells is small, making them to be close.

Aedes koreicus

SUPSI

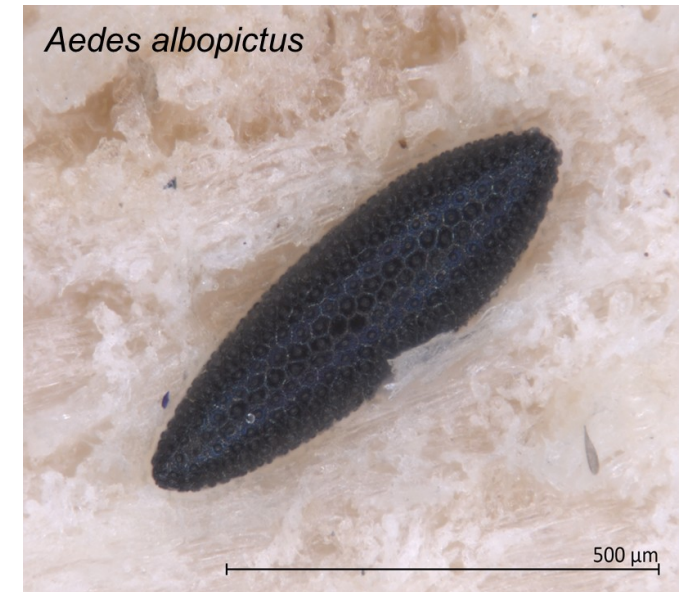
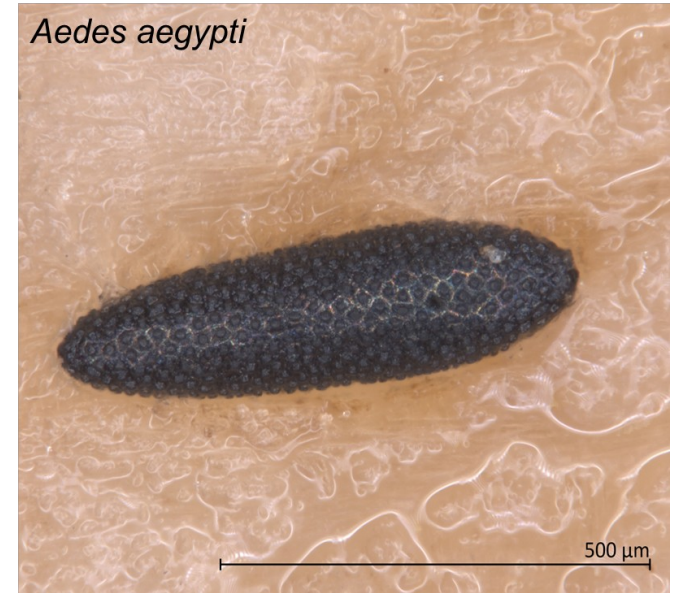


The **outer chorionic cell** consist of a small group of 2-3 long tubercles.

The **exochorionic network** that is found between outer chorionic cells is large, making them to be distant.

EGGS IDENTIFICATION: chorion optical identification

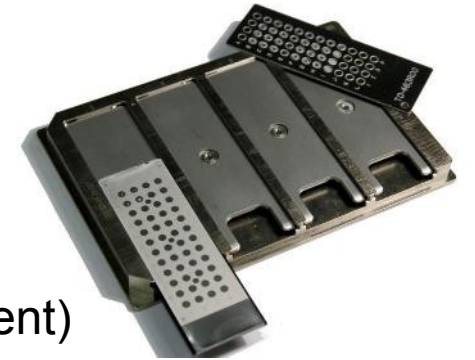
- Both species have a very similar egg chorion structure and the micropillar ring was not easy to see, contrary to what Bova et al. (2016) suggested.
- We are organizing further field samplings in order to confirm the feasibility of the optical tool.



EGGS IDENTIFICATION

SUPSI

mabritec



MALDI-TOT MS t.

- Expensive
- Sample preparation (egg non-reusable)
- Takes time to get a result (preparation + device)
- Budget-limited analyses (small part compared to all the eggs present)
- Fresh (max 3 weeks) and not hatched
- Molecularly accurate (not always 100%)

Optical id.

- Quite cheap
- No sample preparation (egg reusable)
- Results in few minutes (15 min)
- Identification of every egg present (if you can see the chorion structure)
- Images tend to be blurry after a while
- Continuous adjustments to have satisfying pictures (light, focus)

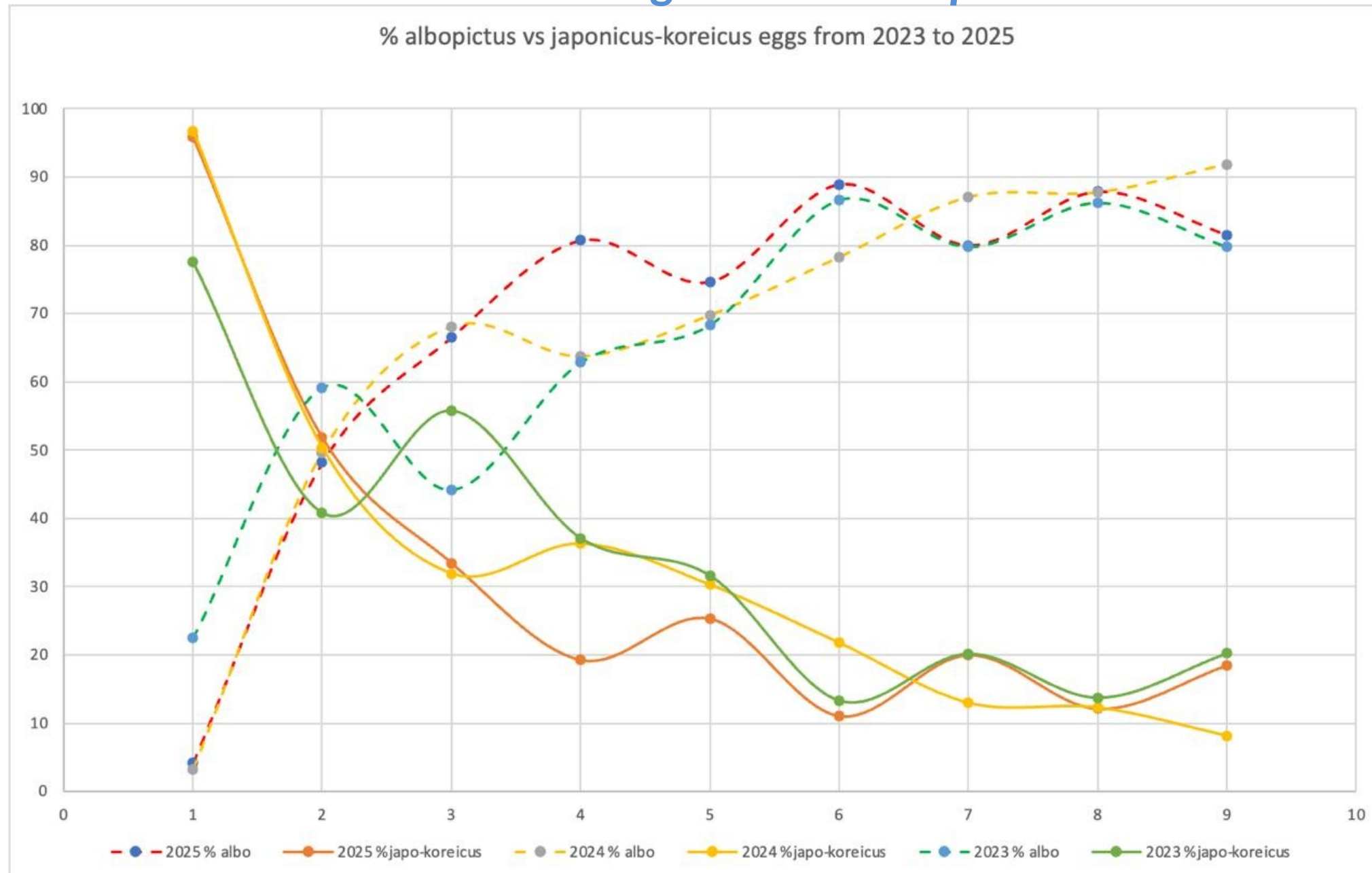


EGGS IDENTIFICATION

- Presence: *Ae. albopictus*, *Ae. japonicus*, *Ae. koreicus* and *Ae. geniculatus*
- *Ae. aegypti*?
- 7 min eggs counting; 15 minutes optical id. + counting

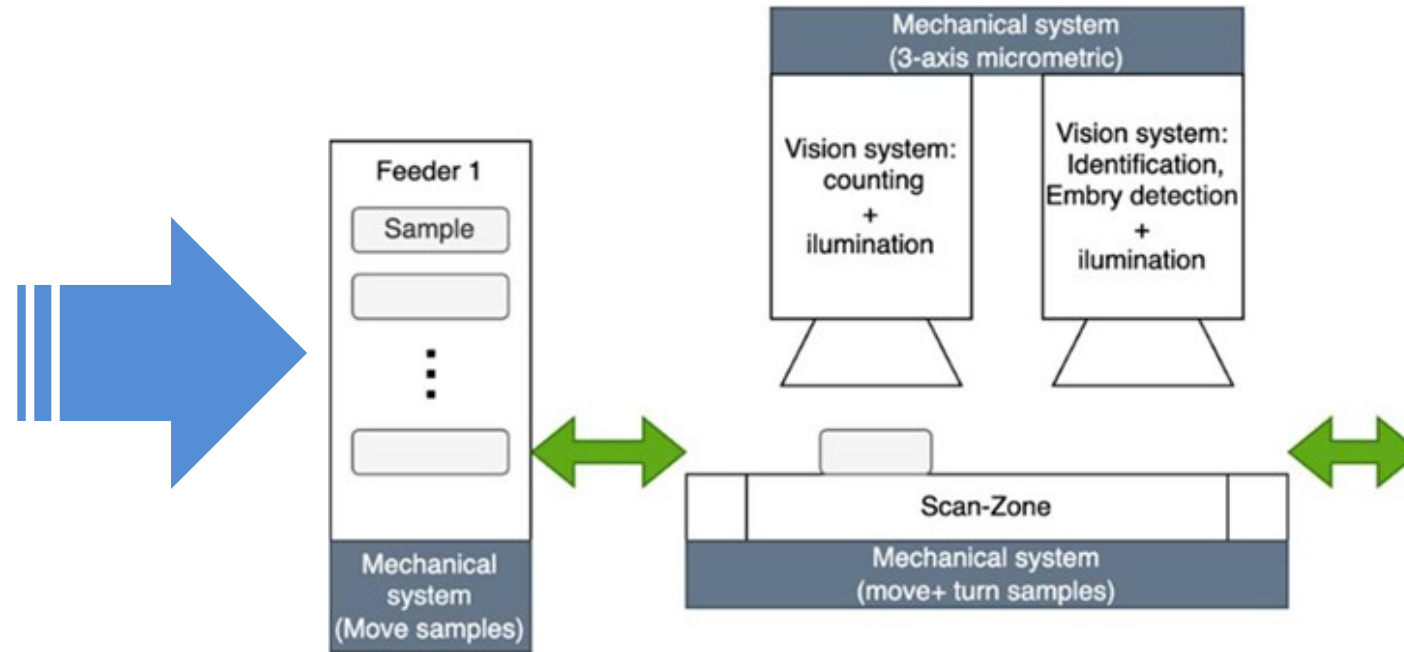
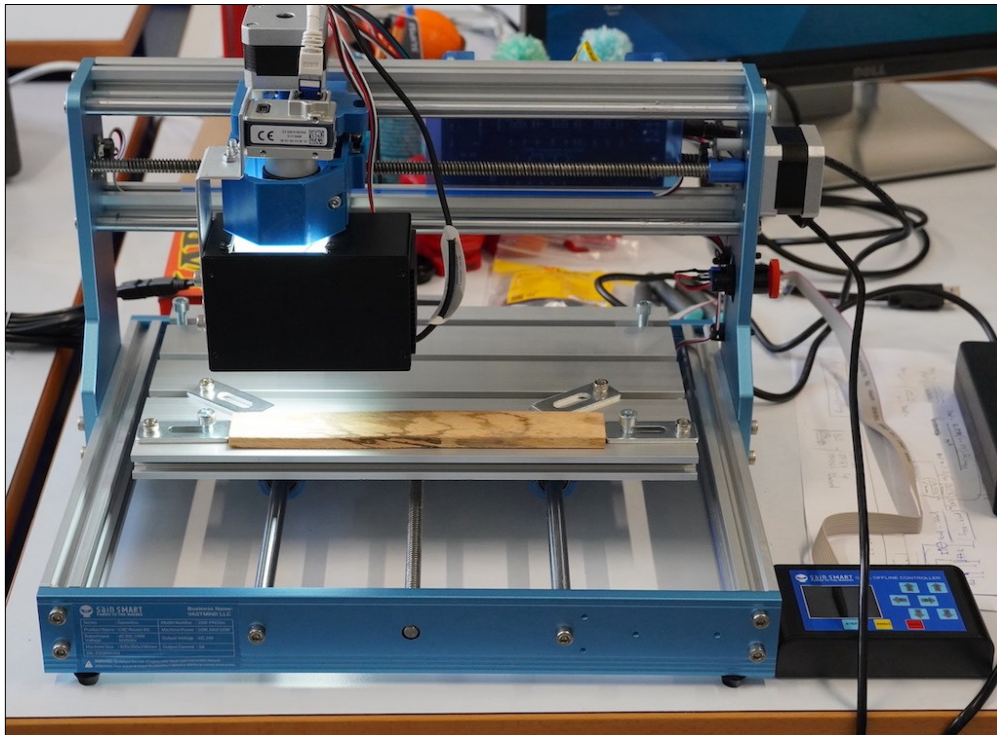


EGGS IDENTIFICATION: estimating real *Ae. albopictus* numbers?



EGGS IDENTIFICATION: proof-of-concept system developed

SUPSI



Project in collaboration with

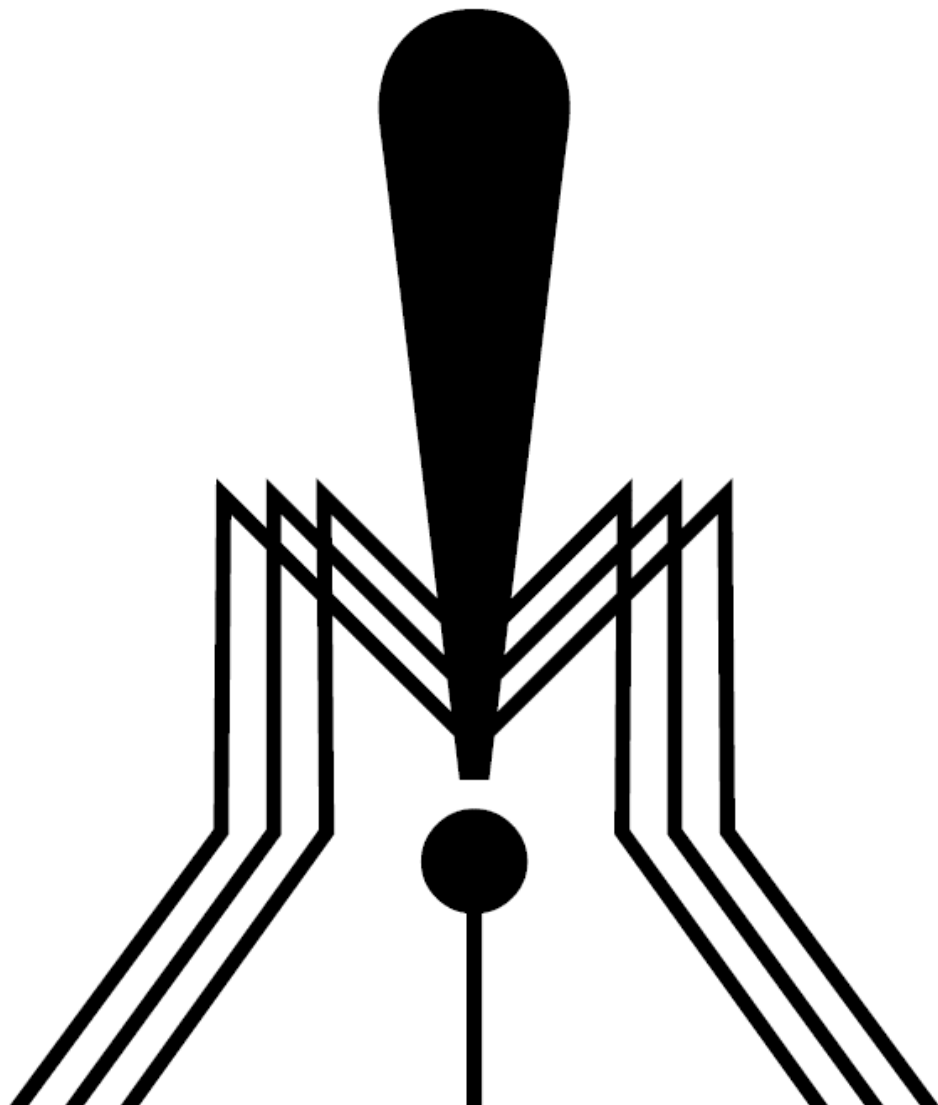
2020

Armando Rivero Salazar, Christian Valles, Roberto Gardenghi and Daniele Allegri

Photonics lab, Department of Innovative Technologies (DTI), SUPSI-DTI

2027

- Three invasive species are spreading across Europe: *Ae. albopictus*, *Ae. japonicus*, and *Ae. koreicus*
- *Ae. aegypti* is also likely to be present
- Ovitrap are the most suitable and least expensive tools for extensive surveillance systems
- At present, the data on ovitraps are not accurate
- Morphological identification works for *Ae. albopictus*
- However, an automatic machine is needed to obtain accurate data and lower costs
- We are trying ...



ti  Repubblica e Cantone Ticino
Dipartimento della sanità e della socialità

ti  Repubblica e Cantone Ticino
Dipartimento del territorio

 Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Bundesamt für Umwelt BAFU

 Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Eidgenössisches Departement des Innern EDI
Bundesamt für Gesundheit BAG

GRAZIE!